Chu Kai Le

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Education

National University of Singapore (NUS) Aug 2026 – May 2029
B.Eng. Mechanical Engineering, Second Major in Computing *Singapore*

- **Awards:** CDE Engineering Scholars Programme (Full-ride, non-bonded Scholarship)
- **Relevant Coursework:** CS2040DE Data Structures and Algorithms

NUS High School of Mathematics and Science Jan 2018 – Nov 2023
NUS High Diploma (High Distinction) *Singapore*

- **Major CAP:** Mathematics: 4.9/5.0, Chemistry: 4.9/5.0, Physics: 4.6/5.0

Experience

Genk Capital Feb 2026 – May 2026
Quantitative Research Intern *Singapore*

- Derived a microstructure signal for price-level collapses in large-tick assets; signal delivered a consistent **+19pp lift** in collapse persistence across all flow regimes (**40%** → **59%** in the tradable regime), validated over **106 days** of L2 LOB data.
- Modeled passive spread-capture entries using a **three-outcome framework** (fill, adverse selection, or flat); demonstrated positive net EV across all queue positions after accounting for one-way taker fees.
- Conducted **survival analysis** on **605K L2 updates** to validate a **97ms median time-to-collapse**, confirming the signal was operationally viable for real-time execution.
- Ran parameter grid sweeps across regimes and instruments; identified regime-dependent fill probability as the primary EV sensitivity driver and systematically excluded suboptimal signal configurations.

Projects/Competitions

Walk-Forward Analysis Engine | *Python, NumPy, Numba*

- Built a walk-forward pairs-trading research framework incorporating rolling in-sample/out-of-sample validation, Kalman-filter hedge ratios, stationarity testing, and transaction-cost modelling.
- Benchmarked NumPy and Numba implementations of path-dependent backtesting to evaluate computational performance across repeated simulations.

Crypto Trading Research (Personal Project) | *Python*

- Ran market-making bots with small real-money positions on crypto and US-stock perpetuals; diagnosed why live trading lost money despite profitable backtests: quotes went stale faster than they could be cancelled, so **94%** of static-quote fills were adverse.
- Recorded order-book data for **330+ instruments** across multiple exchanges and backtested trading ideas under out-of-sample, fee and concentration checks; rejected nearly all of them.

IMC Prosperity 04 (2026) | *Python*

- Placed **123rd / 18,803** globally in university algorithmic trading competition; led strategy for a 5-person team of IMO/IOI/IChO gold medalists.

Technical Skills/Awards

Programming: Python (Pandas, NumPy, Numba), SQL, VBA, Bash
Systems & Tools: Linux, Git, SSH, real-time data pipelines
Quantitative Methods: Statistical modelling, survival analysis, backtesting, time-series analysis
Awards: Singapore Armed Forces National Serviceman of the Year (2026)
Interests: Valorant (Peak Top 0.8%), Trackmania (Peak Top 1 SG, Top 1% Asia), Bouldering (V7), Motorcycling